Retail Sales Data Cleaning & Validation

Microsoft Excel | Portfolio Project

1. Project Overview

This project demonstrates the process of cleaning, validating, and preparing a retail sales dataset for analysis using Microsoft Excel. The raw dataset contained 12,100 rows and 20 columns and included missing values, duplicate records, inconsistent formats, invalid numerical values, and business-rule issues.

2. Project Objective

The objective was to transform the raw dataset into a reliable, consistent, and analysis-ready dataset while preserving valid information and documenting the decisions made during the cleaning process.

3. Dataset Preparation

A backup copy of the original worksheet was created before cleaning. The dataset was reviewed using Excel filters, sorting, formulas, and validation checks.

4. Data Quality Issues Identified and Addressed

Area	Action Taken
Duplicate records	Removed 100 duplicate rows, leaving 12,000 records for the cleaned dataset.
Missing Age	Calculated a median age of 41 and used it to fill missing Age values.
Missing Email	Missing email values were retained because a reliable email address could not be inferred.
Discount	Standardized discount values to a consistent percentage format and corrected cells stored or displayed inconsistently.
Quantity	Identified and corrected negative quantity values and removed fully blank transaction rows.
Unit Price / Cost	Checked for negative values and corrected invalid negative entries.
Profit	Investigated negative profit values and recalculated or validated profit using the underlying sales and cost information.
Order Date / Ship Date	Converted text dates into valid Excel dates using Text to Columns with MDY formatting.
Date validation	Created a helper validation check to identify transactions where the shipping date occurred before the order date.
Order ID	Investigated highlighted duplicate Order IDs and confirmed that the repeated Order ID represented different line items rather than automatically treating them as duplicate transactions.
Phone Number	Standardized Nigerian phone numbers to the +234 international format while preserving blank values.

5. Validation Approach

After cleaning, the dataset was reviewed using filters and helper checks to confirm that key numerical columns no longer contained unintended negative quantities or costs, dates were stored consistently, phone numbers followed the intended +234 format, and duplicate records were investigated rather than removed blindly.

6. Tools and Excel Techniques

Microsoft Excel; Excel Tables; Sort & Filter; Conditional Formatting; Text to Columns; IF; LEFT; RIGHT; TRIM; PROPER; MEDIAN; Paste Special; helper columns; and data validation checks.

7. Outcome

The final dataset was prepared for downstream analysis and reporting. The project demonstrates practical skills in data profiling, data cleaning, standardization, validation, and business-rule checking using Excel.

8. Portfolio Note

The accompanying workbook contains the raw and cleaned worksheets. Before-and-after screenshots are included separately to provide a quick visual demonstration of the transformation process.